5.1.2 Running PostgreSQL as a container

For the Catalog Service application, we'll use a relational database, PostgreSQL, to store the data about the books in the catalog (www.postgresql.org). PostgreSQL is a popular open source database with strong reliability, robustness, and performance, supporting both relational and non-relational data. Most cloud providers offer PostgreSQL as a managed service, freeing you from dealing with issues like high availability, resilience, and persistent storage on your own. Examples are Azure Database for PostgreSQL, Amazon RDS for PostgreSQL, Google Cloud SQL for PostgreSQL, Alibaba Cloud ApsaraDB RDS for PostgreSQL, and DigitalOcean PostgreSQL.

Later in the book, we'll deploy the Polar Bookshop system to a Kubernetes cluster managed by a cloud provider, and I'll show you how to use their offerings for managed PostgreSQL. You'll want to ensure environment parity, as recommended by the 15-Factor methodology, so you'll use PostgreSQL in development as well. Docker makes running databases locally easier than ever, so I'll show you how to run PostgreSQL as a container on your local machine.

In chapter 2, you gave Docker a first try with the Catalog Service application. Running PostgreSQL as a container is no different. Make sure your Docker Engine is up and running, open a Terminal window, and execute the following command:

```
$ docker run -d \
    --name polar-postgres \                ❶
    -e POSTGRES_USER=user \                ❷
    -e POSTGRES_PASSWORD=password \        ❸
    -e POSTGRES_DB=polardb_catalog \       ❹
    -p 5432:5432 \                         ❺
    postgres:14.4                          ❻
```